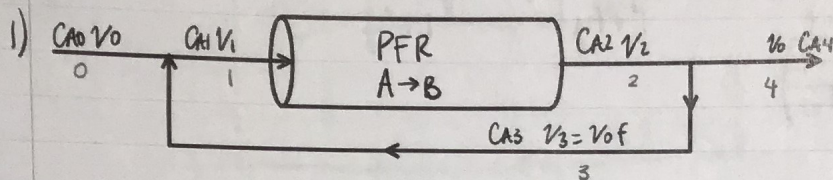


Arwan

Homework 3

Saturday, February 8, 2020 7:18 PM

100



$$X_T = \frac{F_{A0} - F_{A4}}{F_{A0}} = 1 - \frac{F_{A4}}{F_{A0}} = 1 - \frac{C_{A4} V_0}{C_{A0} V_0} = 1 - \frac{C_{A4}}{C_{A0}} = 1 - \frac{C_{A1}(1-X)}{C_{A0}} = 1 - \frac{C_{A0}(1-X)}{C_{A0}(1+Xf)} = 1 - \frac{1-X}{1+Xf}$$

$$C_{A4} = C_{A1}(1-X) = C_{A2} = C_{A3}$$

$$V_0 C_{A0} + f V_0 C_{A3} = V_0(1+f) C_{A1} \rightarrow C_{A0} + C_{A1} f(1-X) = C_{A1}(1+f)$$

$$C_{A1} = \frac{C_{A0}}{1+Xf}$$

PFR: $\frac{dF_A}{dV} = r_A$

$$F_A = F_{A1}(1-X) = C_{A1} V_1(1-X) = \frac{C_{A0}}{1+Xf} V_0(1+f)(1-X)$$

$$r_A = -K C_A = -K C_{A1}(1-X) = -K C_{A0} \left(\frac{1-X}{1+Xf} \right)$$

$$C_{A0} V_0(1+f) \frac{d}{dV} \left[\frac{1-X}{1+Xf} \right] = -K C_{A0} \left[\frac{1-X}{1+Xf} \right] \rightarrow \frac{d}{dV} \left[\frac{1-X}{1+Xf} \right] = \frac{-K}{V_0} \left[\frac{1-X}{(1+Xf)(1+f)} \right]$$

$$\frac{-(1+Xf) - f(1-X)}{(1+Xf)^2} \frac{dX}{dV} = \frac{-K}{V_0} \left[\frac{1-X}{(1+Xf)(1+f)} \right] \rightarrow \frac{+(1+f)}{(1+Xf)^2} dX = \frac{+K}{V_0} \left[\frac{1-X}{(1+Xf)(1+f)} \right] dV$$

$$\int \frac{dX}{(1-X)(1+Xf)} = \frac{K}{(1+f)^2} \int \frac{dV}{V_0}$$

$$\frac{1}{(1-X)(1+Xf)} = \frac{A}{1-X} + \frac{B}{1+Xf}$$

$$1 = A(1+Xf) + B(1-X)$$

$$X=1 \rightarrow 1 = A(1+f) \rightarrow A = \frac{1}{1+f}$$

$$X=-\frac{1}{f} \rightarrow 1 = B(1+\frac{1}{f}) \rightarrow B = \frac{f}{1+f}$$

$$\frac{1}{1+f} \ln \left[\frac{1+Xf}{1-X} \right] = \frac{KV}{V_0(1+f)^2}$$

$$\ln \left[\frac{1+Xf}{1-X} \right] = \frac{KV}{V_0(1+f)} \rightarrow \frac{1+Xf}{1-X} = \exp \left[\frac{KV}{V_0(1+f)} \right] \rightarrow 1+Xf = (1-X)y \rightarrow 1+Xf = y - Xy$$

$$1-y = -Xy - Xf \rightarrow X = \frac{y-1}{y+f}$$

$$X = \frac{e^{KV/V_0(1+f)} - 1}{e^{KV/V_0(1+f)} + f}$$

$$X_T = 1 - 1-X = 1 - \frac{e^{KV/V_0(1+f)} - 1}{e^{KV/V_0(1+f)} + f} = \frac{1+f - e^{KV/V_0(1+f)} + 1}{1+f} = \frac{2+f - e^{KV/V_0(1+f)}}{1+f}$$

$$X_T = 1 - \frac{1-X}{1+Xf} = \frac{1 - \frac{e^{KV/v_0(1+f)} - 1}{e^{KV/v_0(1+f)} + f}}{1 + f \left(\frac{e^{KV/v_0(1+f)} - 1}{e^{KV/v_0(1+f)} + f} \right)} = \frac{1 + fy - 1 + y}{1 + fy} = \frac{y(1+f)}{1 + fy}$$

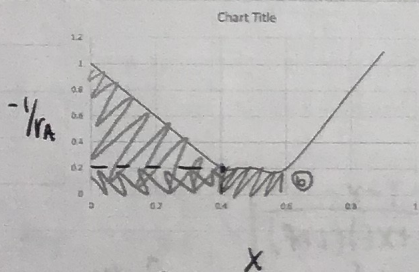
$$X_T = \frac{\frac{e^{KV/v_0(1+f)} - 1}{e^{KV/v_0(1+f)} + f} (1+f)}{1 + f \left(\frac{e^{KV/v_0(1+f)} - 1}{e^{KV/v_0(1+f)} + f} \right)}$$

2)

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
-r _A (mol/L.min)	1	1.67	5	5	5	5	1.25	0.91

F_{A0} = 300 mol/min A → B + C

a) PFR & CSTR Volumes for X = .4



$$V_{PFR} = 300[(.4)(.2) + (.4)(.8)(.5)] \rightarrow 72L = V_{PFR}$$

$$V_{CSTR} = 300(.4)(.2) \rightarrow 24L = V_{CSTR}$$

b) 0.4 ≤ X ≤ 0.9 V_{CSTR} = V_{PFR}

The graph shows that the area under the curve is the same as X times y or conversion times -1/r_A.